

on the Jetson Orin NX: Python, NumPy, OpenCV, no deep-learning framework

on the flight controller: Betaflight 4.4.3

30 fps

~28 fps, 47 ms packet age

4 RC channels at 40 Hz

camera

IMX477, 20 deg up-tilt
1920x1080 at 30 fps
HFOV 74 deg, fx 425 px

gate detector

YOLOv8n-pose hand497
640x360, 27 ms on GPU
8 corners, kp conf 0.25

observation

55 ch x 32 frames = 1760
latched 30 Hz, delayed 1-4
tracker picks the next gate

policy

NumPy MLP
3 x 256, 40 Hz
thrust + 3 rates

MSP override

mask 15 on AUX5
no timeout on
the last frame

**PID loop,
motors**

440 deg/s at
full stick
DShot300 x4

telemetry back: attitude, raw gyro, status and RC in one MSP_MULTIPLE_MSP request,
p95 20.3 ms with the stream running, on the same 115200-baud UART

adapter on the command path: inverts the rate and throttle curves, pitch sign -1, gyro counts / 16.384

the radio pilot, who holds the only abort

arming, and the override switch: the abort path

transmitter

AUX1 = ARM (radio only)
AUX5 down = policy flies
AUX2 = ANGLE

props-off handover check

streams neutral sticks first (the buffer
starts at zero, which reads as failsafe),
then proves AUX5 off returns control

the missing sensor

no usable height: the barometer collapses
under prop wash, so height comes
from the gate in view